



University
of Idaho

ASSIGNMENT 4A: PROGRESS REPORT

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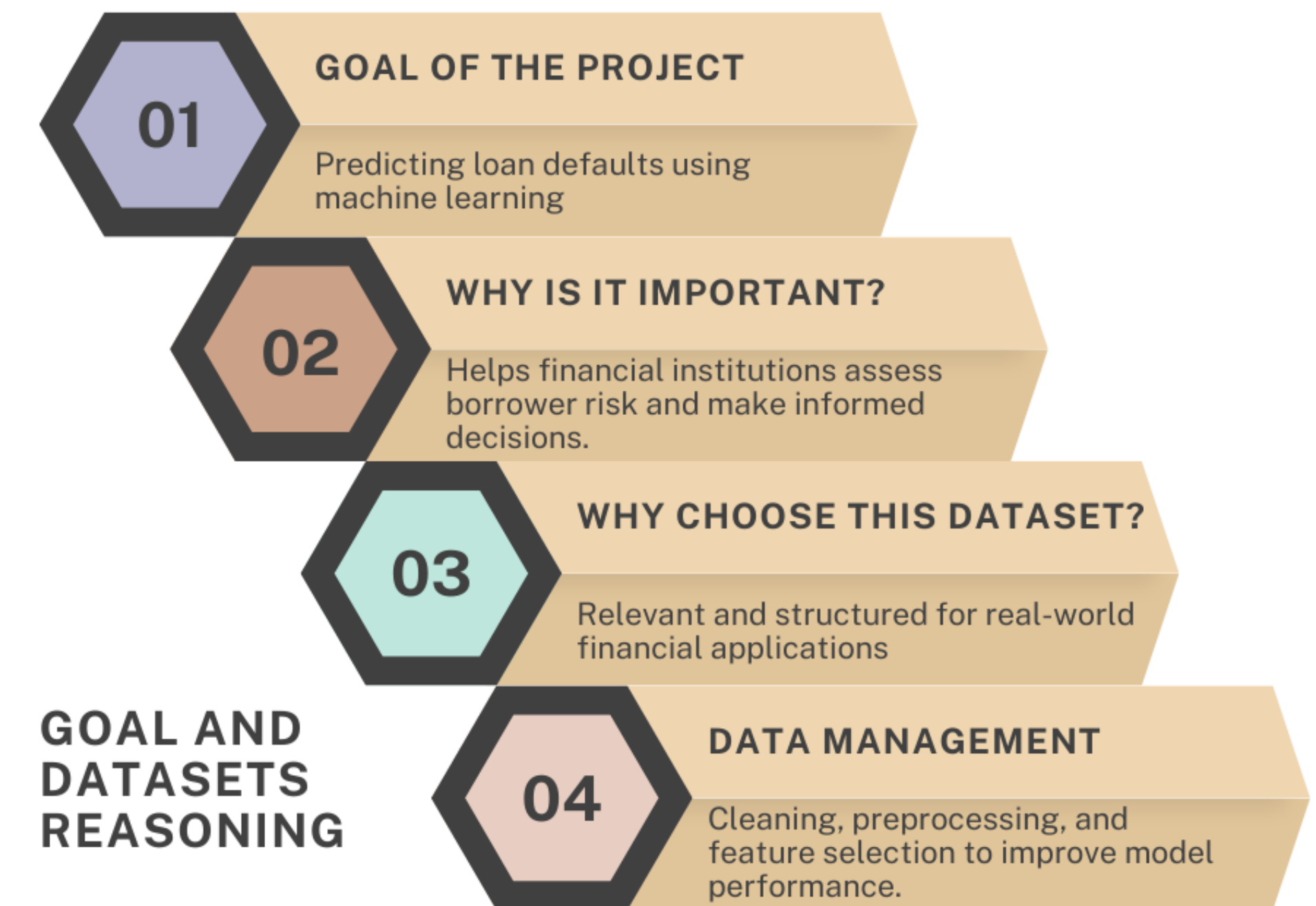
*** SUMIT SHAHI**

GOAL AND MODE OF COLLECTION



Initially, we considered working with a stock-related dataset but struggled to define a clear objective. Instead, we chose to analyze loan default predictions, as it provides a well-defined problem with real-world financial implications.

- The **goal of this project** is aimed at predicting the likelihood of loan default.
- **Why It's Important** – Helps financial institutions assess borrower risk and make informed decisions.
- **Why This Dataset?**
 - Large dataset (255,347 rows, 18 columns).
 - Sourced from Coursera's Loan Default Prediction Challenge.
 - Relevant and structured for real-world financial applications.
- **Data Management** – Cleaning, preprocessing, and feature selection to improve model performance.



A **loan default** occurs when a borrower fails to meet the terms of a loan agreement, typically by missing payments or violating other conditions. This prediction is equally important for financial institutions when making loan approval or denial decisions, as it allows them to assess the risk of default based on various features.

DATA FORMAT & DATASET STRUCTURE



I Dataset Name: Loan Default Prediction Dataset

- **File Format:** Loan_default.csv (CSV format)
- **Size:** 255,347 rows × 18 columns (24.83 MB)

I Type of Data:

- **Numerical:** Age, Income, CreditScore, etc.
- **Boolean:** HasMortgage, HasCoSigner, HasDependents
- **Categorical/text:** Education, EmploymentType, MaritalStatus, LoanPurpose, etc.

METADATA & STANDARDS



I Metadata = “data about data” (e.g., column type and meaning)

- This dataset includes structured **JSON metadata**
- **Data Provenance:** Metadata includes links to Kaggle, dataset creator information, and a licensing statement.

I Standards Used:

- **Schema.org** – defines types like Text, Integer, Boolean
- **MLCommons Croissant** – for machine-readable schema

I Other fields include: name, dataType, and description

I Example:

```
1  {  
2    "name": "CreditScore",  
3    "dataType": "Integer",  
4    "description": "The credit score of the borrower"  
5  }  
6  
7
```

DATA MANAGEMENT

DATA MANAGEMENT PLAN USING CATEGORIES

- I Creation of Logical Collections:** The dataset consists of borrower records that can be categorized into logical groups—demographics, finances, credit history, and loan amount.
- I Physical Data Handling:** The dataset is in CSV format. We're using version control via GitHub, and backup to OneDrive/Google Drive. Replication across team environments ensures integrity and efficient model training.
- I Interoperability Support:** Our project integrates the Kaggle dataset with potential additional sources such as credit bureau data. Data formats such as CSV and JSON ensure compatibility with statistical Software like Python (Pandas, Matplotlib) and R. Metadata also aligns with the standards provided my schema.org.

DATA MANAGEMENT

DATA MANAGEMENT PLAN USING CATEGORIES

- I Security Support:** Access to data is strictly limited to team members with permission. We use GitHub's privacy settings to ensure that borrower information remains secure. To build trust in our loan default predictions, we also document every action and verify all changes made to the data.
- I Data Ownership:** The original dataset is owned and maintained by Coursera. We downloaded the copy hosted by Kaggle. Our team ensures data quality and cleaning, with the leader overseeing interpretation. This guarantees reliable models and accurate, well-informed decisions.
- I Metadata Collection, Management, and Access:** The metadata for the dataset included column descriptions, data provenance, and change history. Additionally, we aim to implement maps that show how different pieces of data are related to each other.

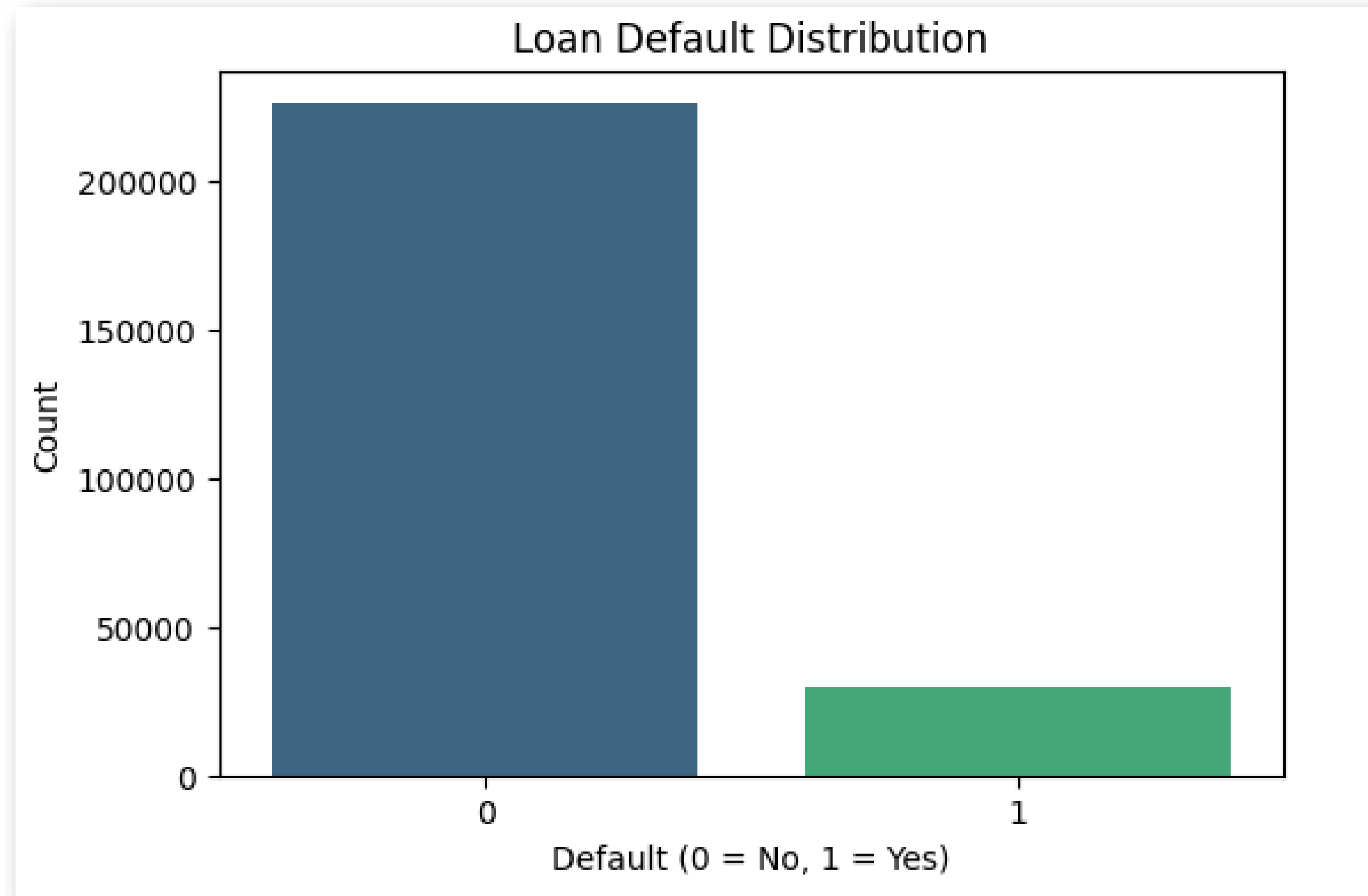
DATA MANAGEMENT

DATA MANAGEMENT PLAN USING CATEGORIES

- I Persistence:** The loan default dataset will be retained indefinitely for historical analysis, with procedures for data migration as technologies change. We plan to use host the Dataset, Metadata and the analysis in GitHub to reduce risk, and regular reviews keep data and analysis relevant.
- I Discovery:** We will provide clear GitHub documentation with transparent README files, allowing team members and collaborators to efficiently find specific borrower attributes for analysis. This documentation serves as a valuable resource for both new and current members, ensuring a clear understanding of the data.
- I Data Dissemination and Publication:** We plan to notify users of changes through GitHub updates and documentation, while protecting sensitive borrower data. We will regularly update our datasets on GitHub to support collaboration and analysis replication.

GitHub: https://github.com/AankitPaudel/Final_Project-Data_Science-479

INITIAL DATA PRE-PROCESSING AND EDA



INITIAL DATA PRE-PROCESSING AND EDA



```
3 data.info()
```

Summary of DataFrame:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 255347 entries, 0 to 255346
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   LoanID                255347 non-null object
1   Age                  255347 non-null int64
2   Income               255347 non-null int64
3   LoanAmount           255347 non-null int64
4   CreditScore          255347 non-null int64
5   MonthsEmployed       255347 non-null int64
6   NumCreditLines        255347 non-null int64
7   InterestRate         255347 non-null float64
8   LoanTerm              255347 non-null int64
9   DTIRatio             255347 non-null float64
10  Education             255347 non-null object
11  EmploymentType        255347 non-null object
12  MaritalStatus         255347 non-null object
13  HasMortgage           255347 non-null object
14  HasDependents         255347 non-null object
15  LoanPurpose           255347 non-null object
16  HasCoSigner           255347 non-null object
17  Default               255347 non-null int64
dtypes: float64(2), int64(8), object(8)
memory usage: 35.1+ MB
```

```
1 data.isnull().sum()
```

LoanID	0
Age	0
Income	0
LoanAmount	0
CreditScore	0
MonthsEmployed	0
NumCreditLines	0
InterestRate	0
LoanTerm	0
DTIRatio	0
Education	0
EmploymentType	0
MaritalStatus	0
HasMortgage	0
HasDependents	0
LoanPurpose	0
HasCoSigner	0
Default	0
dtype:	int64

INITIAL DATA PRE-PROCESSING AND EDA



```
3 data.describe()
```

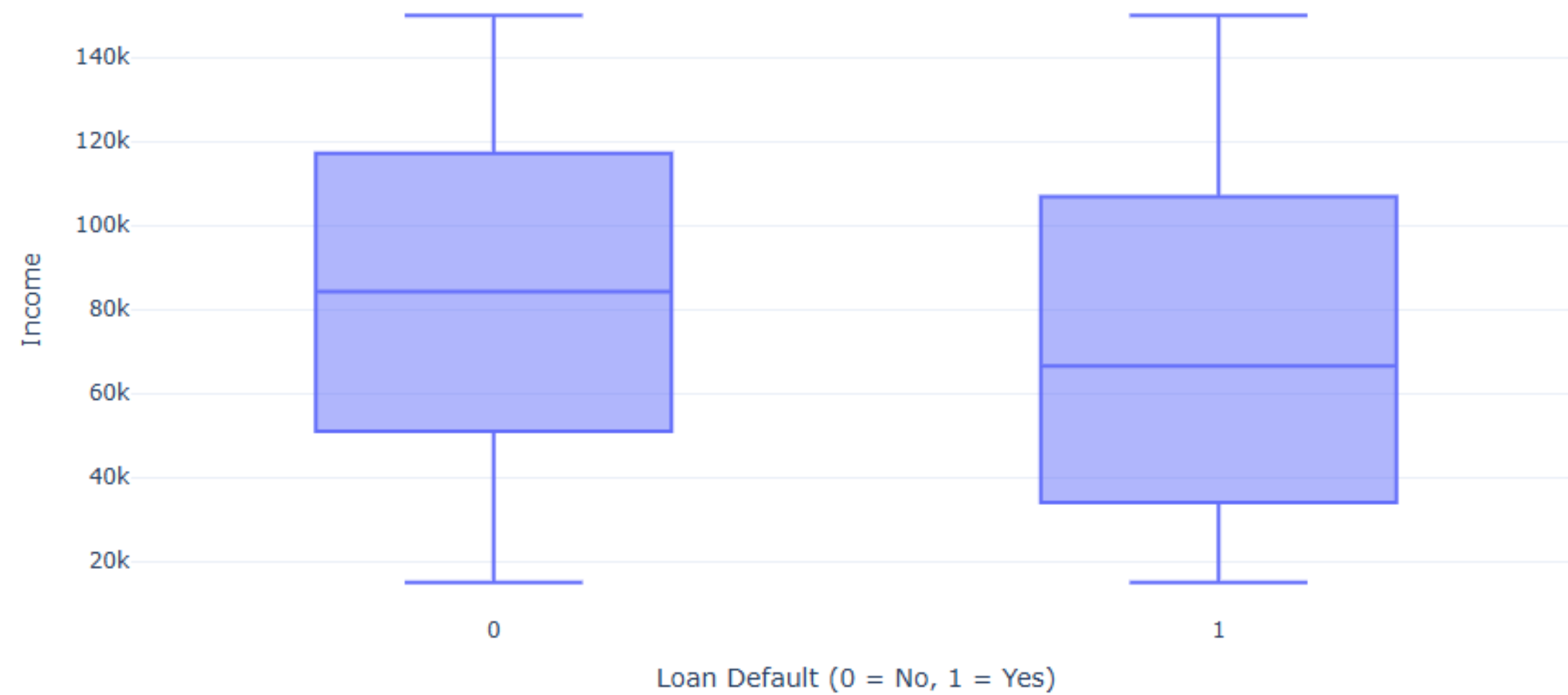
Descriptive Statistics of the DataFrame:

	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio
count	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000
mean	43.498306	82499.304597	127578.865512	574.264346	59.541976	2.501036	13.492773	36.025894	0.500212
std	14.990258	38963.013729	70840.706142	158.903867	34.643376	1.117018	6.636443	16.969330	0.230917
min	18.000000	15000.000000	5000.000000	300.000000	0.000000	1.000000	2.000000	12.000000	0.100000
25%	31.000000	48825.500000	66156.000000	437.000000	30.000000	2.000000	7.770000	24.000000	0.300000
50%	43.000000	82466.000000	127556.000000	574.000000	60.000000	2.000000	13.460000	36.000000	0.500000
75%	56.000000	116219.000000	188985.000000	712.000000	90.000000	3.000000	19.250000	48.000000	0.700000
max	69.000000	149999.000000	249999.000000	849.000000	119.000000	4.000000	25.000000	60.000000	0.900000

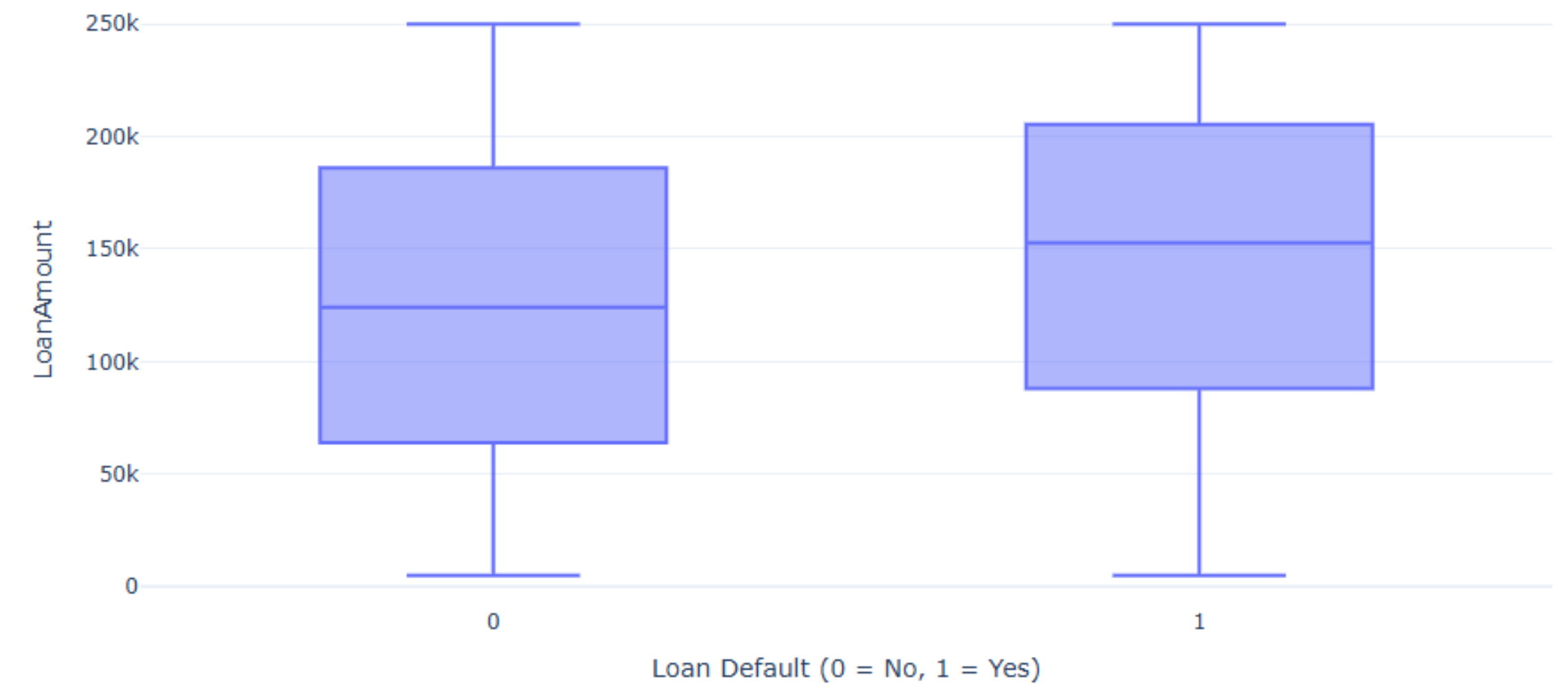
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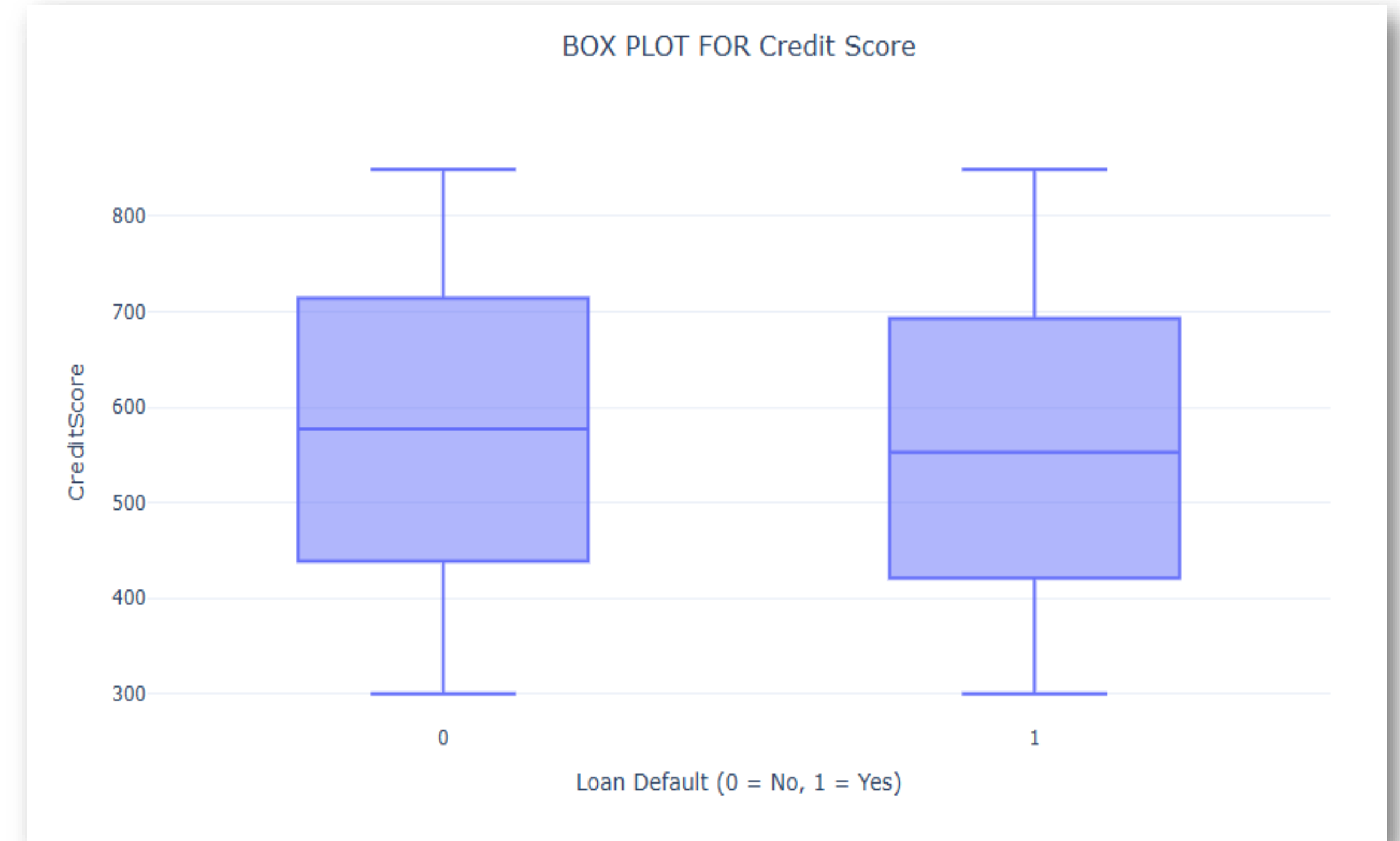
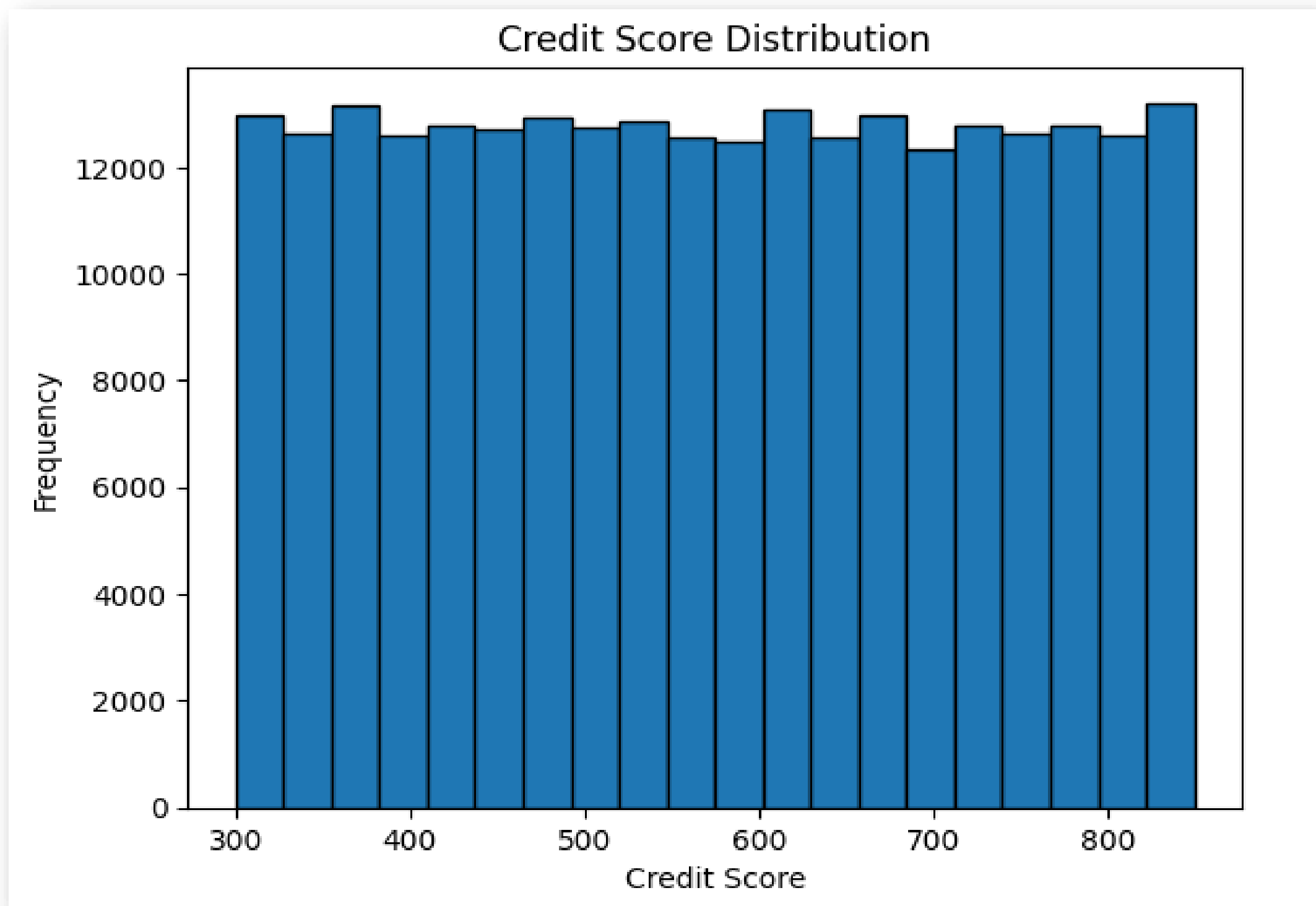
BOX PLOT FOR Income



BOX PLOT FOR LoanAmount



INITIAL DATA PRE-PROCESSING AND EDA



REFERENCE:



- Data Science Coding Challenge: Loan Default Prediction*. Coursera. (n.d.).
<https://www.coursera.org/projects/data-science-coding-challenge-loan-default-prediction>
- Nikhil. (2023, September 11). *Loan default prediction dataset*. Kaggle.
<https://www.kaggle.com/datasets/nikhil1e9/loan-default/data>